Machine Learning Models – Important Parameters (Including Bagging & Boosting)

Quick revision notes for important machine learning models and their key parameters that data scientists commonly use in projects and interviews.

Linear Regression

Parameter	Meaning
fit_intercept	Whether to calculate intercept
copy_X	Copy input data or overwrite
n_jobs	Number of CPU cores used

Logistic Regression

Parameter	Meaning
penalty	Regularization type (l1, l2, elasticnet)
C	Inverse regularization strength
solver	Optimization algorithm
max_iter	Maximum iterations
class_weight	Handle imbalanced data

K-Nearest Neighbors (KNN)

Parameter	Meaning
n_neighbors	Number of neighbors
weights	Uniform or distance weighting
algorithm	Search algorithm
metric	Distance metric

Decision Tree

Parameter	Meaning
criterion	Split quality (gini or entropy)
max_depth	Maximum tree depth
min_samples_split	Minimum samples to split

min_samples_leaf	Minimum samples in leaf
max_features	Features used for split

Bagging Classifier

Parameter	Meaning
n_estimators	Number of base models
max_samples	Samples drawn for each model
max_features	Features drawn for each model
bootstrap	Sampling with replacement

Random Forest

Parameter	Meaning
n_estimators	Number of trees
max_depth	Maximum tree depth
min_samples_split	Minimum samples to split
min_samples_leaf	Minimum leaf samples
max_features	Features considered for split

Extra Trees (Extremely Randomized Trees)

Parameter	Meaning
n_estimators	Number of trees
max_depth	Maximum depth
min_samples_split	Minimum split samples
min_samples_leaf	Minimum samples in leaf
max_features	Number of features used

AdaBoost

Parameter	Meaning
n_estimators	Number of weak learners
learning_rate	Contribution of each learner
algorithm	Boosting algorithm type

Gradient Boosting

Parameter	Meaning
n_estimators	Number of boosting stages
learning_rate	Step size shrinkage
max_depth	Depth of individual trees
subsample	Fraction of samples used

XGBoost

Parameter	Meaning
n_estimators	Number of trees
learning_rate	Boosting learning rate
max_depth	Maximum tree depth
subsample	Fraction of training samples
colsample_bytree	Features used per tree

LightGBM

Parameter	Meaning
num_leaves	Maximum leaves per tree
learning_rate	Boosting learning rate
n_estimators	Number of boosting iterations
max_depth	Maximum depth
feature_fraction	Fraction of features used

CatBoost

Parameter	Meaning
iterations	Number of trees
learning_rate	Learning rate
depth	Tree depth
l2_leaf_reg	L2 regularization